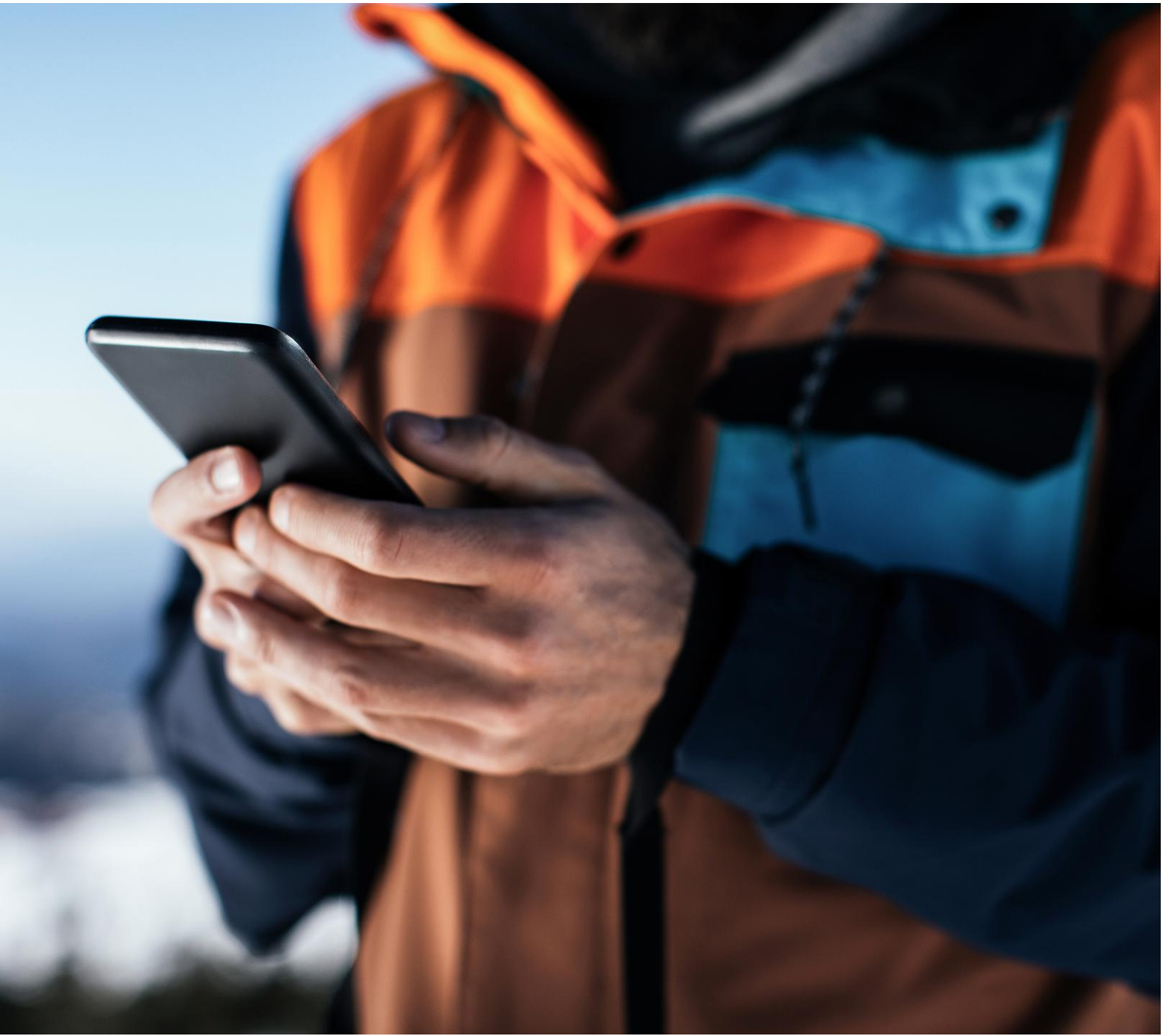


Smartphone Movement Data in Skiing

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GNSS data in skiing

Smartphones with integrated GNSS functionality make it possible to track snowsports movement data in real time. The data provides valuable insights into riding behaviour and could contribute to improving safety on the slopes in future. However, this comparative study shows that the degree of accuracy depends on the type of device, the environment and the skiing speeds.

Starting point

Nowadays, smartphones are much more than just a means of communication. With integrated GNSS (Global Navigation Satellite System) functionality, they are increasingly being used for real-time positioning, navigation and tracking. GNSS combines signals from several satellite systems such as the American GPS and the European Galileo system. Many apps use this functionality to capture the smartphone's current location and display it in the app.

Such GNSS-based apps are also widely used for skiing and snowboarding and offer a variety of features: they measure speeds, travelled distances and altitude differences, provide orientation or offer entertainment through virtual competitions. As a result, these apps generate a wealth of positioning and time data and can provide insights into skiers' movement behaviour.

Some ski resorts are already using the information gleaned from apps to plan and improve their services. This allows them, for instance, to better calculate lift capacities or manage visitor flow in a more targeted manner. Some ski resorts collect and use data independently by making their own apps available to visitors.

The movement data generated by apps is also of great interest in terms of accident prevention. This makes it possible to detect patterns that indicate an elevated accident risk, such as skiing speeds or usage frequency on certain sections of the slope. Little research has been conducted into these parameters until now, but in future they could help to identify hazardous areas on the slopes and mitigate them with targeted measures. In road traffic, similar data is already being used by fitness apps such as Strava Metro to analyse the behaviour and safety of cyclists [1]. Such a method could also be applied in snowsports in the medium term.

Data quality is critical when using skiers' and snowboarders' movement data to improve safety on the slopes, because the exact position and speed are key to conducting meaningful analyses.

Objective

This study compares the positioning and speed data of four apps and four smartphone models with a reference system to ascertain their accuracy, thus enabling a substantiated assessment of the benefits and limitations of smartphone movement data.

Methodology

Study design

In the winter of 2022/23, smartphone GNSS data was collected in the ski resorts of Davos and Zermatt. Two skiers were equipped with four smartphones in total: two Android and two iOS devices, each comprising a high-end and a low-end model. All devices had the same four apps installed to record the positioning and time data (iSKI, Skitude, Slopes, Strava). Both skiers were also equipped with a differential GNSS positioning system (DGNSS). Thanks to the correction via a base station, this system can accurately measure to a range of less than 10 cm and served as a benchmark to check the accuracy of the smartphones and apps. More details of the study methodology can be found in the publication by Petrella et al. [2].

Data collection and analysis

The DGNSS reference data was recorded and stored at a frequency of 50 Hz, the smartphone app data at a frequency of 1 Hz.

The first five minutes of the recording were excluded from the analysis, as major errors often occur in this phase, before the received satellite signals have been optimised. As the vertical component of the positioning data is strongly influenced by the underlying elevation model, only the horizontal positioning data was used. This is referred to in the following as “positioning data”. Only downhill data was taken into account.

The speed (km/h) was derived from the exported positioning data. To this end, the distance between two recorded positions was determined and divided by the time between two position measurements. This calculation from the raw data is necessary because apps use different calculation methods and automatically generated speed data cannot be compared like for like.

Determining accuracy

The Euclidean distance – i.e. the straight-line distance between the respective positions and the corresponding time-synchronized DGNSS reference data – was calculated for each smartphone/app combination. This allowed the position and speed error to be determined in relation to the reference system.

Findings

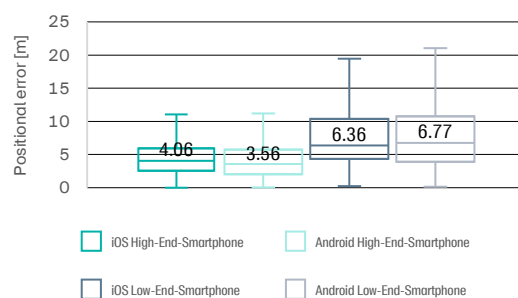
Greater positioning precision with high-end smartphones – no difference between apps

The smartphones had an average positional error of 4.5 m. Despite excluding outliers (based on the three-standard deviation requirement), the positional error ranged up to 25 m, briefly exceeding 100 m in individual cases. The latter can be detected and excluded by checking the speed. At three quarters of the measuring points, the positional error was less than 7 m.

The high-end iOS and Android smartphones delivered the most reliable positioning data, with a positional error of around 4 m (Figure 1). In comparison, the two low-end smartphones showed a wider average positional error of around 6 m and a greater

dispersion of values. The difference between the individual smartphone models was statistically significant, with the greatest effect being observed between high-end Android and iOS devices and the corresponding low-end versions. The examined apps had no effect on positioning accuracy.

Figure 1: Comparison of positional errors between high-end and low-end smartphones



Using positioning data to reliably determine speed

The average speed error across all phones was less than 2 km/h, and three quarters of all measurement points showed an error of less than 3.5 km/h. There were no differences between high-end and low-end smartphones and no significant differences in the calculated speed across the different apps.

Accuracy depends on speed travelled

The positional errors increased significantly at high travelling speeds. The average positional error at slow travelling speeds (up to 14 km) was around 3 m, at higher travelling speeds (~50 km) it was more than 5 m. The speed error was lowest in the 22–36 km/h range. The error increased at slower or faster speeds and, at 4.5 km/h, was highest at speeds of around 65 km/h. This speed-dependent data accuracy is well known and also occurs with other position detection systems [3].

Accuracy depends on the environment

The positional error was dependent on the location in which the measurement was taken. There are two

main reasons for this well-known phenomenon: one factor is the number of available satellites, the other is the characteristics of the terrain and the environment, which can block or reflect satellite signals. In Davos, the wide terrain favoured measurement accuracy. In the Zermatt region, which is surrounded to the south by high mountains, the positional error was around 2.5 m wider, while the speed errors did not show any significant difference.

Conclusion – what are potential uses for smartphone movement data?

The study shows that the accuracy of smartphone movement data can vary greatly and is influenced by several factors, including:

- the smartphone model
- the environment and the associated number of available satellites
- the travelling speed

Movement data must be specifically selected for accident prevention purposes in snowsports. Positional errors persist even with large data volumes.

- Positioning data from smartphones can be used to evaluate parameters that do not depend on the precise location of skiers – for instance traffic volume on a slope, usage frequency and usage density.
- Positioning data from smartphones is not suitable for evaluating parameters that depend on the exact position, e.g. choice of trajectory, utilisation of the slope width and near collisions. This requires the use of higher precision systems such as DGNSS.

Alongside positioning data, the derived speeds can also be used for accident prevention purposes.

- Smartphone speed data is suitable for estimating the travelling speed of slope users, provided it is calculated from the raw position data. Speed data generated automatically by apps is not usually comparable like for like.

In future, increasing smartphone performance and improved measurement technology could lead to higher data accuracy. However, accuracies of less than one metre are unlikely in dynamic applications such as skiing.

Recommendations for data collection with smartphones

- 1. Unobstructed view of the sky**
GNSS signals can be interfered with by objects such as buildings, trees or mountains. In dense forests or narrow valleys, signal quality may be significantly impaired and lead to larger positional errors. For this reason, measurements should be taken in open terrain wherever possible; failing this, topographical restrictions should be taken into account when interpreting the data.
- 2. High-quality smartphones provide more accurate measurements**
Newer and higher-quality smartphones tend to measure more accurately and provide more reliable positioning data.
- 3. Wait for a stable connection**
At the start of a measurement, allowing around 5 minutes to establish a stable connection from the smartphone to the satellites is recommended. The first 5 minutes of each measurement should be excluded from the data analysis.
- 4. Use only clean raw data**
Raw data should be used to perform meaningful analyses. The data should be cleaned with standardised procedures to allow data from different apps to be combined.
- 5. Speed affects positional accuracy**
Fast skiing speeds can affect positional accuracy. This means that data accuracy should be treated with caution at high speeds.
- 6. Keep an eye on app updates**
Several apps can be used simultaneously on one smartphone, as the position is stored centrally and accessed by the apps. When installing updates, pay attention to possible modifications such as new smoothing methods which may affect the raw data.

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Recommendations for data collection with smartphones



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